

Toxicity Detection - Literature Review and Project Plan

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March 21, 2025

1 Preprocessing and Words' Representation

English, Finnish, and German differ significantly in grammar and morphology, which affects NLP models' cross-lingual performance. English is an analytic language with relatively simple morphology (lacking cases and rich inflection) [1], whereas Finnish is agglutinative with extensive case marking and complex inflections, and German is fusional with moderate inflection (e.g. four cases, grammatical gender). These structural disparities influence how well a model trained on English can generalize to the others [2]. For example, German is linguistically closer to English (sharing many cognates and syntactic traits) [3], so an English-trained model may transfer slightly better to German than to Finnish, which is from a distant family and highly morphologically rich. Recent analyses of multilingual BERT (mBERT) show that differences in morphosyntax correlate with zero-shot transfer performance [2] – larger linguistic gaps (e.g. English vs. Finnish) lead to lower transfer accuracy.

Another challenge is that malicious users often introduce adversarial character perturbations to evade toxicity filters. Minor edits like character swaps, insertions, deletions, or using look-alike symbols (homoglyphs) can preserve a word's meaning to a human but confuse a model [4]. For instance, inserting punctuation into an offensive word (“i.d.i.o.t”) or replacing letters with similar characters can bypass naive detectors. Such simple obfuscations have been shown to degrade classifier performance – authors in [4] attained a 98% attack success rate against a state-of-the-art toxicity detection model using small word/character tweaks.

This section focuses on academia's view on how to normalize and tokenize text across languages, handle adversarial noising, and choose effective embeddings (from word-level to contextual multilingual transformers) to maximize cross-lingual generalization.

1.1 Preprocessing

1.1.1 Cleaning

Toxic comment corpora often contain noisy user-generated text. A common first step is to normalize casing, spelling, and formatting. For example, most studies lowercase text (when case is not informative) and remove irrelevancies (e.g. HTML codes, extra whitespace, or artifacts). Standard punctuation is usually kept (since “!?” , “**”, etc. might convey intensity or sarcasm) [5]. It might also be beneficial to address repeated characters.

This cleaning improves efficiency and prevents vocabulary explosion due to useless tokens. Since the text samples addressed in this survey are expected to be from social media platforms, some additional preprocessing is needed. When normalizing offensive content, it is often necessary to handle leetspeak and deliberate obfuscations: for example, mapping zero “0” to letter “o”, “@” to “a”, or stripping punctuation inserted inside words [6]. The authors in [7] presented a RNN based word recognizer to mitigate the effects of this. Additionally, since social media platforms usually contain parole (spoken language), it can be beneficial to expand contractions such as “don't” to “do not”.

1.1.2 Translation

Depending on the chosen classifier, the training data may need to be translated to the target languages. Models trained on multilingual data may be able to generalize to other languages as well. However, in [8], this was shown to be less effective in the case of Finnish, when compared to machine translation alternative.

1.2 Representation

After basic cleaning, the text must be tokenized. Older pipelines used whitespace or language-specific rules to split words (Finnish compound words might be split into components). In modern NLP applications, subword tokenization is widely used to handle multilingual vocabularies. Algorithms like WordPiece[9], SentencePiece[10], or Byte-Pair Encoding (BPE) break words into subword units, which is especially good for morphologically complex languages. Subword tokenization mitigates the out-of-vocabulary problem by allowing rare or unseen words to be composed from known pieces. For example, a Finnish insult formed by agglutinating a suffix can still be recognized via its root subword.

2 Base Model’s Architecture

In the context of toxicity detection, establishing a strong baseline is important for evaluating the effectiveness of more complex models. For this task, we propose using Logistic Regression or Random Forests. These methods are suitable for binary classification tasks such as separating toxic and non-toxic comments, and they provide a solid foundation for comparing the performance with the main models.

2.1 Logistic Regression

Logistic Regression is a linear model commonly used in text classification tasks due to its simplicity and strong performance [11]. It works by modeling the probability of a comment belonging to a specific class, such as toxic or non-toxic, using a logistic function. The input features for Logistic Regression are typically derived from text representations like Bag-of-Words (BoW) or TF-IDF, which capture the frequency or importance of words in the text [12].

One of the key strengths of Logistic Regression is its interpretability. The model’s coefficients can be directly analyzed to understand how each feature contributes to the classification decision. This makes it particularly useful for tasks like toxicity detection, where understanding the influence of specific words or phrases can provide valuable insights. Additionally, Logistic Regression is computationally efficient and performs well on small to medium-sized datasets, making it a practical choice for initial experiments or as a baseline [12].

2.2 Random Forest

Random Forest is another effective baseline model for toxicity classification. It consists of an ensemble of decision trees trained on different subsets of the dataset, reducing overfitting and enhancing the generalization. Random Forest can capture nonlinear patterns in the text, making it more flexible for complex linguistic structures. Prior research has shown that Random Forest excels in scenarios where rule-based classification is beneficial, such as detecting accusatory or targeted language in online discussions [13].

A decision tree-based approach is particularly well-suited for toxicity classification because it systematically selects features that maximize information gain. The model constructs a hierarchy of classification rules, where each node in the tree represents a decision based on an n-gram or another textual feature. This step-wise approach helps in quantifying how strongly a particular feature contributes to classifying a comment as toxic or non-toxic [13].

3 Modeling

3.1 Brief

Toxicity detection in its simplest form can be seen as a binary classification problem coupled with text processing. That is a very general framework that has been extensively studied in the past, and a lot of different solutions have been proposed. Moving forward, we will focus on machine learning methods and leave other considerations for future work.

3.2 Traditional machine learning

Traditional machine learning refers to so-called shallow methods such as logistic regression, support vector machines, and decision trees. These methods rely on features extracted from text using some rule-based approaches. Features can include bag-of-words, lexicon-based, part-of-speech, or any arbitrary features [14]. The classifier then uses those features to make predictions. Although this is a widely used approach due to its good efficiency and somewhat sufficient results, modern research is leaning toward deep learning methods [15], which simplify the process by learning good representations without hand-made features.

3.3 Deep learning

Deep learning is based on different artificial neural network strategies that learn important patterns from text devising representations that are used to make the final classification. When a lot of data is available the superior performance of these algorithms over traditional machine learning has been shown over and over again [11]. In the text classification context, popular models include recurrent neural networks and convolutional neural networks [14].

Traditionally word representations learned by these models have been static, meaning that same words have same representations independent of the context or partially contextual as a result of the nature of the models. As research on embeddings has advanced scholars have proposed better contextual representation strategies such as bidirectional recurrent neural networks [16]. These contextual representations have proven to be extremely powerful since they are able to encode information depending on the context on both sides of the words. This idea of contextual representations combined with the notion of the attention first used with encoder-decoder recurrent neural networks for machine translation [17] have led us to the next topic that’s being revolutionizing field of the natural language processing since 2017.

3.4 Transformers

Ever since the introduction of the transformer architecture [18] the whole field of natural language processing has been constantly changing, and studies have been more focused on different transformer architectures as they have yielded excellent results in downstream tasks such as toxicity detection [5].

The original transformer model was created for machine translation task and therefore consists of encoder-decoder architecture supporting the conditional generation objective. The success of causal decoder only models have been also demonstrated on different downstream tasks on several occasions [19, 20].

Since causal language models usually require immoderately large number of parameters, we restrict ourselves to encoder only models also known as masked language models, which were quickly discovered after the initial emergence of transformers.

The original masked language model BERT [21] and several of its descendants were trained on English data, which doesn’t directly support multilingual task. Fortunately, researchers have realized the importance of multilingualism and several models with similar architectures have been trained on multilingual data [22, 23] allowing models to process textual data on multiple languages in shared space. This has been shown to increase model performance significantly, especially for low resource languages [23].

Transformer models pre-trained with a large amount of the data tend to create biases towards datasets they have seen. Toxicity detection in social media comments can deviate from commonly used pre-training data and therefore some sort of domain adaptation might be needed before fine-tuning final model. There are also off-self models trained with social media data [24], which might alleviate this problem.

4 Cross-lingual Transfer Approaches for Multilingualism

The approaches can be classified into three categories: Instance Transfer, Feature Transfer, and Parameter Transfer [25]. (See Figure 1)

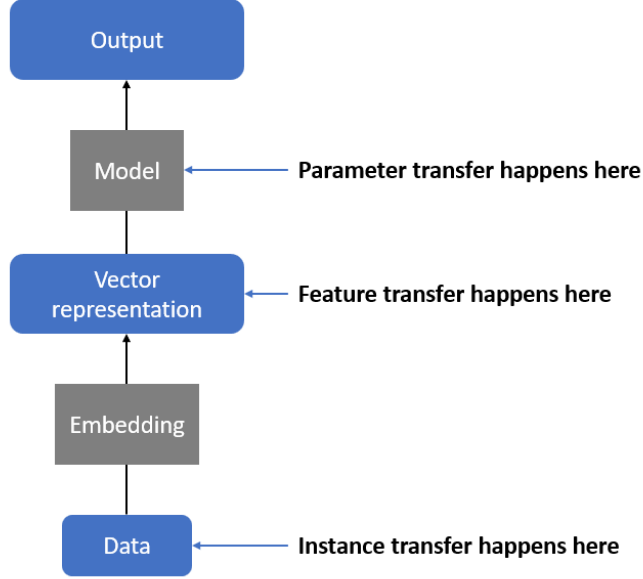


Figure 1: Three transfer scenarios

4.1 Instance Transfer

Instance transfer involves transferring data, including texts and labels, from the source language(s) to the target language(s) [main]. The result of this is a dataset of translated or aligned instances between the source and target languages, enabling the training of a classifier in the target language(s) without the need for manually annotating data. This can be achieved by machine translation (text-based instance transfer) [26, 27], pseudo labeling (label-based instance transfer) [27], leveraging parallel corpus [28], or hybrid methods [27].

4.2 Feature Transfer

Feature Transfer operates on the latent feature space, transferring the data’s underlying features [25]. The result of this is a joint vector space that both source and target languages can be projected on (words with the same meaning have close representations), allowing the model to generalize patterns learned from the source language to the target language. Some widely used methods include Multilingual Distributional Representations [29], which treat identical words in different contexts as the same (e.g., "bat" in baseball vs. the animal). In contrast, Multilingual Contextual Representations [30] adapt to different contexts, distinguishing between word meanings based on usage.

4.3 Parameter Transfer

Parameter Transfer occurs at the model parameter level, where the model’s behavior is transferred from the source to the target language. There are three main training scenarios in transfer learning. In zero-shot transfer, the model is trained only on the source language(s) and directly applied to the target language(s) without additional training [31]. In joint learning, the model is trained on both source and target languages at the same time, allowing it to learn shared patterns [32]. In cascade learning, the model is first trained on the source language(s) and then fine-tuned on the target language(s) [33].

5 Project Plan

5.1 Data Strategy

5.1.1 Data Sources

- We are going to use the data provided by the course.

5.1.2 Preprocessing

- We remove URLs, special characters, and HTML tags.
- We convert emojis to text or tokens.

5.2 Feature Engineering

5.2.1 Text Representation

- For the LSTM, we use GloVe or FastText for embeddings.
- For the XLM-R model, there is no tokenization needed, since it uses SentencePiece.

5.3 Model Architecture

5.3.1 Baseline Models

- We use a vanilla LSTM as the baseline model.

5.3.2 Main Models

- We fine-tune XLM-R on English and perform zero-shot transfer to German and Finnish.

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